

Field Leaf Image Collection Protocol

Jensina Davis · Original protocol: September 2025

Standalone illustrated edition: September 7, 2026

Collect three leaf images per plot using the imaging box and FieldBook. This document includes the setup, collection, navigation and export instructions demonstrated in the accompanying recordings; no videos are needed to follow it.

1. Prepare the equipment

1. Clip the neck strap to the box hooks using the carabiners. Adjust the strap to a comfortable working height. To change its length, remove the Velcro near the carabiner, undo the metal twist tie, adjust the doubled rope, and reattach the tie and Velcro.
2. Secure the phone on top with the Velcro straps. Align its rear camera with the opening so it faces into the box.
3. Install three AA batteries in the light battery pack. Press its button to switch on the lights. Check that the LED strip is inside the box, above the lip/foam.
4. Fully charge the phone and any power bank before collection. The original protocol reports approximately four hours of active collection per full phone charge; plan charging accordingly.

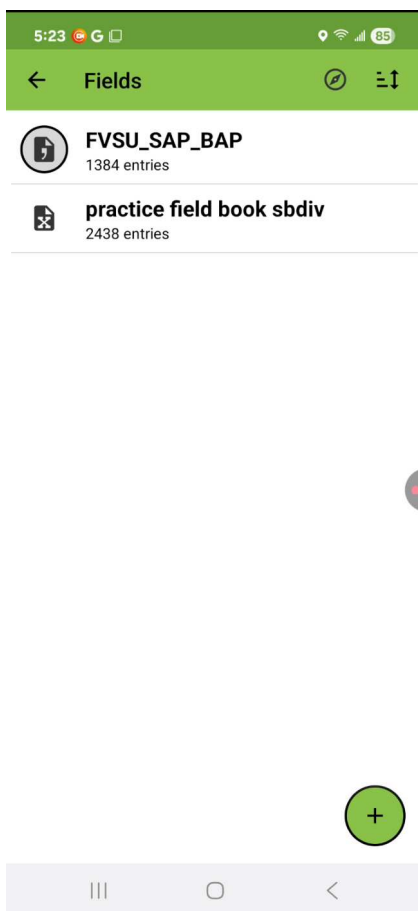


Prepared imaging box, phone mount and neck strap.

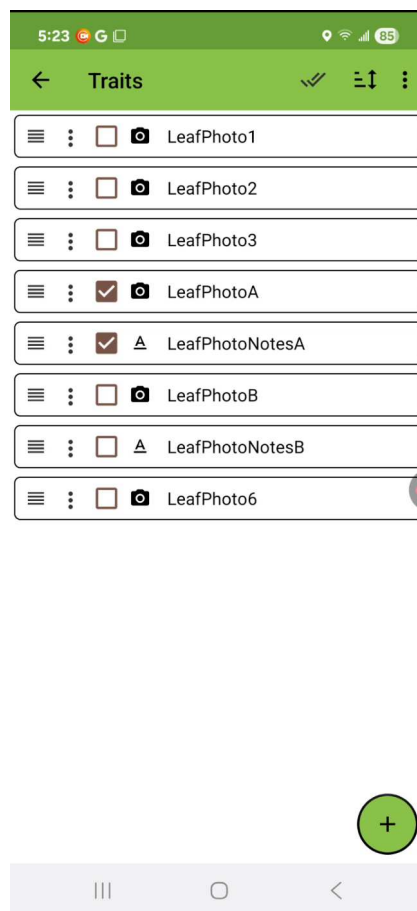
2. Set up FieldBook

1. Open **FieldBook**. From its home screen, open **Fields** and select the field assigned to your experiment.
2. Return to the home screen and open **Traits**. Select the assigned photo trait and its matching notes trait. In the demonstration these are **LeafPhotoA** and **LeafPhotoNotesA**.
3. Return to the home screen and choose **Collect**. Confirm that the population, replication and plot number match the plot you are collecting.

The screenshots show the 2025 demonstration field, **FVSU_SAP_BAP**. Field names, plot numbers and trait names are experiment-specific examples, not settings to reuse for a different experiment.



Select the correct field.

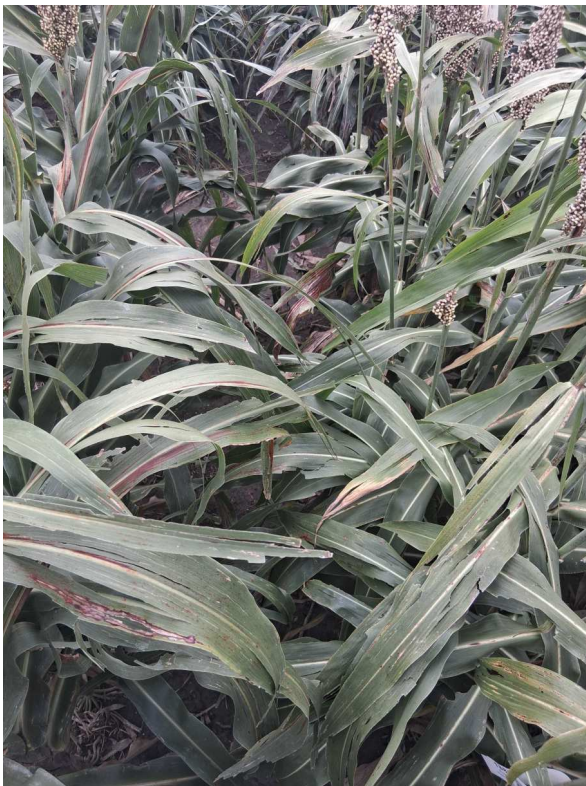


Activate the photo and matching notes traits.

3. Select and position a leaf

Collect three different leaves per plot, one retained photograph per leaf. Check the plot identifier against the field label regularly (at least every 5–10 plots, as in the original protocol), and whenever you jump to another plot.

1. Open the bottom of the box using its handle.
2. Select a representative leaf, avoiding the flag (topmost) leaf, senescing leaves and leaves with large rips. Detach the leaf from the plant. Follow any additional leaf-selection rule assigned for your experiment.
3. Place the leaf on the bottom panel, on the side opposite the reference card and outside the guides, as shown below. Leave the reference card unobstructed.
4. Close the box. Pull the leaf slightly to flatten it before taking the photograph.



Avoid leaves with large rips such as this example.



Place the leaf across the bottom panel, opposite the reference card.

4. Photograph and replace the leaf

1. In **Collect**, select the photo trait. Slide the screen upward if necessary to reveal the camera control, then tap the capture button.
2. Wait for the photograph to appear in the photo strip. Review it before moving on; if it is poor, remove that photograph and retake it (page 5).
3. Open the bottom panel and discard the photographed leaf.
4. Repeat the positioning and capture steps with two more leaves, for **three retained leaf photographs in total for this plot**. Add a note if needed, then advance to the next plot using the black arrows (page 6).



Capture the image with the box closed.



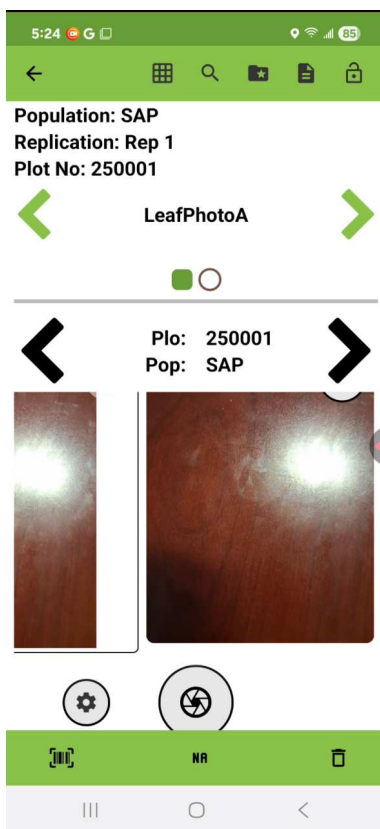
Open the bottom panel and discard the photographed leaf.

5. Review photographs and record notes

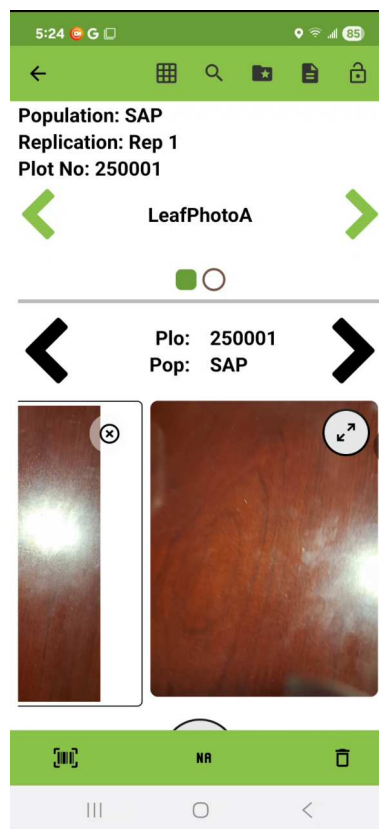
Retake a poor photograph: tap the **X** on that **individual photograph** in the photo strip to delete it, then capture a replacement. Check that the plot still has three usable leaf photographs before leaving it.

Add a note: switch from **LeafPhotoA** to **LeafPhotoNotesA** using the green trait arrows. Type the relevant observation, for example that all available leaves were damaged. Return to the photo trait before photographing another leaf.

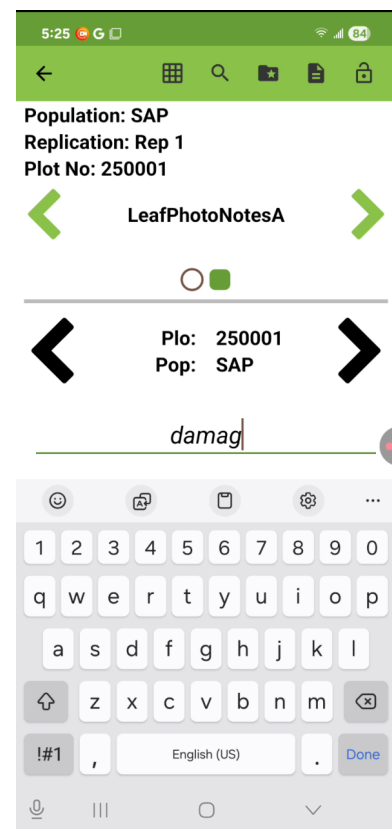
The screens below demonstrate the controls using indoor test photographs; those photographs are not examples of the required leaf images.



Camera control and photo strip.



The X removes an individual photograph.



Enter observations in the matching notes trait.

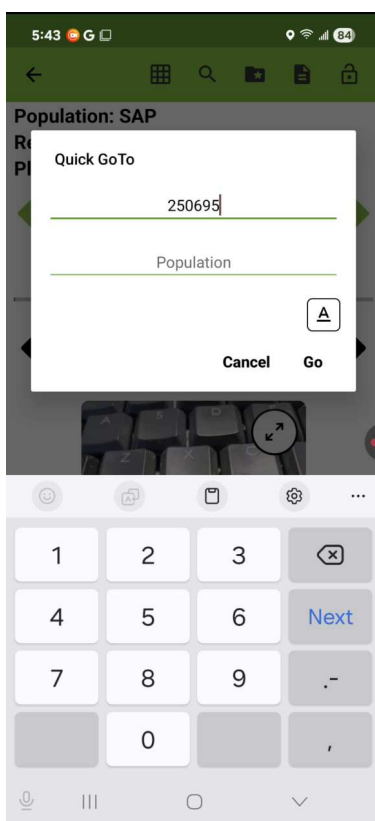
6. Navigate between plots and check records

Next or previous plot: use the **black arrows** beside the plot information. The demonstrated app makes a brief sound when the plot changes. The **green arrows change traits**, not plots.

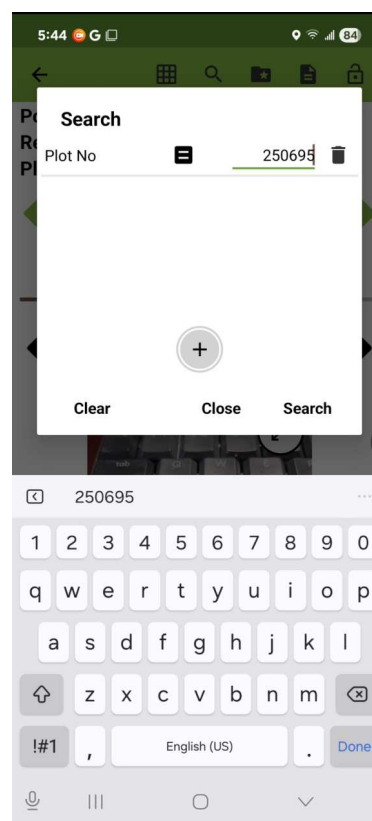
Jump to a known plot: tap the plot information to open **Quick GoTo**, enter the required plot number and tap **Go**. Confirm the displayed plot before collecting.

Search: tap the magnifying-glass icon in Collect. Choose **Plot No**, enter the plot number and tap **Search**. Tap the matching result to open that plot.

Check existing records: tap the data-grid icon in Collect. This view shows data stored on this phone. Recorded photo entries appear as green cells with local file paths; notes appear in their corresponding column. A record on another phone will not necessarily appear here.



Quick GoTo: enter the destination plot.

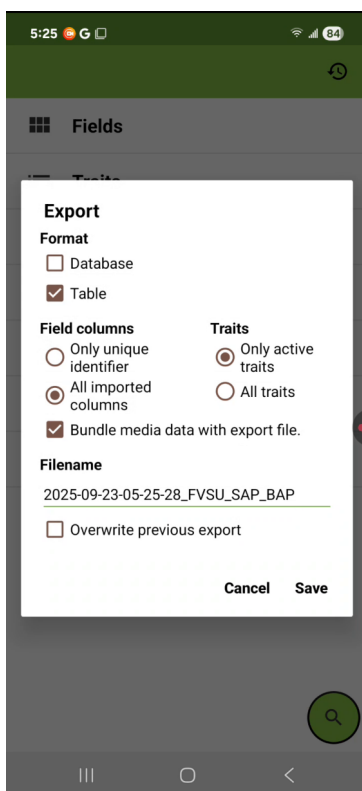


Search for a plot, then select the matching result.

7. Export the photographs with their metadata

At the end of collection, return to the home screen and open **Export**. Use these demonstrated settings:

1. Select **Table** and **All Imported Columns**.
2. Select **Only active traits**, after confirming that both the photo and notes traits are active.
3. Check **Bundle Media Data with Export File**. This is essential for including the photographs.
4. Check the filename. Leave **Overwrite previous export** unchecked to preserve previous exports.
5. Tap **Save** and acknowledge any confirmation dialog. The bundled export is saved as a ZIP file on the phone.



Export settings: include the media, not only the table.

Recommended backup check: copy the ZIP to your project storage and confirm that it opens and contains both the table and photographs. Keep the phone copy until the backup has been checked.

Source note: Original illustrated protocol and three September 2025 demonstration videos. Videos 1–2 supplied the FieldBook instructions; video 3 visually demonstrated leaf handling and contained no intelligible narration. Screenshots retain the demonstrated interface. The backup check is an added safeguard.